INCH-POUND

MIL-C-3098/131C(ER)
24 September 1992
SUPERSEDING
MIL-C-3098/131B(ER)
23 June 1975

MILITARY SPECIFICATION SHEET

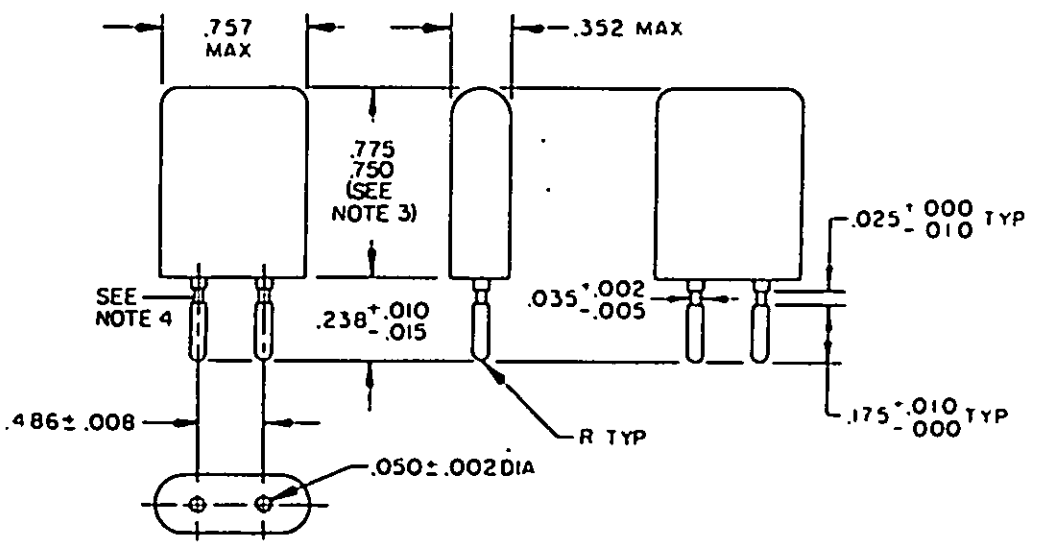
CRYSTAL UNIT, QUARTZ, CR150/U

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

The requirement for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 45 to 100 MHz; fifth mechanical overtone; controlled; series resonance.

Inches	mm
.002	0.05
.008	0.20
.010	0.25
.025	0.64
.035	0.89
.050	1.27
.175	4.45
.238	6.05
.352	8.94
.486	12.34
.750	19.05
.757	19.23
.775	19.69



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Holder is similar to HC-27/U (MIL-H-10056/14) except for minimum limit for height of cover
4. The pin undercut may be omitted.
3. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR150/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: Modified HC-27/U (MIL-H-10056/14).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 45.0 to 100 MHz, inclusive.

Operable temperature range: -55°C to +80°C, inclusive.

Operating temperature range: +70°C to +80°C, inclusive.

Frequency and equivalent resistance:

Equivalent resistance: 40 ohms, maximum.

Reference temperature: 75°C ±1°C.

Frequency tolerance: ±5 ppm.

Mode of oscillator: Fifth mechanical overtone.

Drive level: 1.0 ±0.2 mW.

Capacitance, shunt: 3.0 pF maximum.

Shock (specified pulse):

Frequency change permitted: ±2.5 ppm.

Equivalent resistance change permitted: ±40 ohms, maximum.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±2.5 ppm.

Equivalent resistance change permitted: ±40 ohms, maximum.

Thermal shock:

Frequency change permitted: ±2.5 ppm.

Equivalent resistance change permitted: ±40 ohms, maximum.

Aging:

Frequency change permitted: ±5 ppm.

NOTE: CR-150/U is similar to CR-143/U except for a wider frequency tolerance.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Army - ER

Review activities:
Army - AR, MI
DLA - ES

Preparing activity:
Army - ER

Agent
DLA - ES

(Project 5955-A208-03)

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